

Monomial $f(q) = a \cdot q^9$ Report

Polynomial

$$f(q) = aq^9$$

Naive

$$g_{\text{naive}}(X) = ax^9$$

$$\mathbb{E}[g_{\text{naive}}(X)] = aq^9 + 36aq^7\sigma^2 + 378aq^5\sigma^4 + 1260aq^3\sigma^6 + 945aq\sigma^8$$

$$\begin{aligned}\text{Var}(g_{\text{naive}}(X)) &= 81a^2q^{16}\sigma^2 + 7128a^2q^{14}\sigma^4 + 248724a^2q^{12}\sigma^6 \\ &\quad + 4359096a^2q^{10}\sigma^8 + 40330710a^2q^8\sigma^{10} + 190670760a^2q^6\sigma^{12} \\ &\quad + 411131700a^2q^4\sigma^{14} + 309241800a^2q^2\sigma^{16} + 34459425a^2\sigma^{18}\end{aligned}$$

$$\begin{aligned}\text{MSE}(g_{\text{naive}}) &= 81a^2q^{16}\sigma^2 + 8424a^2q^{14}\sigma^4 + 275940a^2q^{12}\sigma^6 \\ &\quad + 4592700a^2q^{10}\sigma^8 + 41351310a^2q^8\sigma^{10} + 192972780a^2q^6\sigma^{12} \\ &\quad + 413513100a^2q^4\sigma^{14} + 310134825a^2q^2\sigma^{16} + 34459425a^2\sigma^{18}\end{aligned}$$

$$\text{Bias}(g_{\text{naive}}) = 36aq^7\sigma^2 + 378aq^5\sigma^4 + 1260aq^3\sigma^6 + 945aq\sigma^8$$

Unbiased

$$g_{\text{unb}}(X) = 945a\sigma^8x - 1260a\sigma^6x^3 + 378a\sigma^4x^5 - 36a\sigma^2x^7 + ax^9$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^9$$

$$\begin{aligned}\text{Var}(g_{\text{unb}}(X)) &= 81a^2q^{16}\sigma^2 + 2592a^2q^{14}\sigma^4 + 42336a^2q^{12}\sigma^6 \\ &\quad + 381024a^2q^{10}\sigma^8 + 1905120a^2q^8\sigma^{10} + 5080320a^2q^6\sigma^{12} \\ &\quad + 6531840a^2q^4\sigma^{14} + 3265920a^2q^2\sigma^{16} + 362880a^2\sigma^{18}\end{aligned}$$

$$\begin{aligned}\text{MSE}(g_{\text{unb}}) &= 81a^2q^{16}\sigma^2 + 2592a^2q^{14}\sigma^4 + 42336a^2q^{12}\sigma^6 \\ &\quad + 381024a^2q^{10}\sigma^8 + 1905120a^2q^8\sigma^{10} + 5080320a^2q^6\sigma^{12} \\ &\quad + 6531840a^2q^4\sigma^{14} + 3265920a^2q^2\sigma^{16} + 362880a^2\sigma^{18}\end{aligned}$$

Comparison

$$\text{mean gap} = -36aq^7\sigma^2 - 378aq^5\sigma^4 - 1260aq^3\sigma^6 - 945aq\sigma^8$$

$$\begin{aligned}\text{variance gap} = & -4536a^2q^{14}\sigma^4 - 206388a^2q^{12}\sigma^6 - 3978072a^2q^{10}\sigma^8 \\ & - 38425590a^2q^8\sigma^{10} - 185590440a^2q^6\sigma^{12} \\ & - 404599860a^2q^4\sigma^{14} - 305975880a^2q^2\sigma^{16} - 34096545a^2\sigma^{18}\end{aligned}$$

$$\begin{aligned}\text{MSE gap} = & -5832a^2q^{14}\sigma^4 - 233604a^2q^{12}\sigma^6 - 4211676a^2q^{10}\sigma^8 \\ & - 39446190a^2q^8\sigma^{10} - 187892460a^2q^6\sigma^{12} \\ & - 406981260a^2q^4\sigma^{14} - 306868905a^2q^2\sigma^{16} - 34096545a^2\sigma^{18}\end{aligned}$$

$$\begin{aligned}\text{Bias}(g_{\text{naive}})^2 = & 1296a^2q^{14}\sigma^4 + 27216a^2q^{12}\sigma^6 + 233604a^2q^{10}\sigma^8 + 1020600a^2q^8\sigma^{10} \\ & + 2302020a^2q^6\sigma^{12} + 2381400a^2q^4\sigma^{14} + 893025a^2q^2\sigma^{16}\end{aligned}$$